

Hrishabh Sahu

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Work Experience

Graduate Engineering Trainee (GET) at HCL Tech

Aug 2022 - Dec 2023

Worked with the SCCM tool in a US-based insurance company. Responsibilities included deploying MS Windows security patches, software updates to client VDIs, and troubleshooting errors. Received appreciation for outstanding contribution in JFM-23 Quarter.

1-Month Training at ETL Laboratory, Bhilai Steel Plant

Aug 2021 - Nov 2021

Observed the scheduled inspection and maintenance of numerical relays, PLCs, and circuit breakers. Learned about relay testing procedures, troubleshooting methods, and the role of PLCs in industrial automation.

Education

Year	Degree/Certificate	Institute	CPI / %
2025 - 2027	M.Tech / Communication Engg.	Indian Institute of Technology Delhi	7.5/10
2018 - 2022	B.Tech / Electrical & Electronics Engg.	Bhilai Institute of Technology, Durg	8.8/10
2018	Class XII	BSP Senior Secondary School Sector-7, Bhilai	80%
2016	Class X	BSP Senior Secondary School Sector-7, Bhilai	7.6/10

Projects

Implementation of error correcting code (LDPC) in 5G standard.

Nov 2025 (ongoing)

Guide : Prof. Kaushik Saha | Indian Institute of Technology Delhi.

- Currently studying about the Encoder and Decoder algorithms and its implementation techniques.
- Developing pseudocode for LDPC encoder and decoder based on 5G NR specifications.
- Planning software to hardware workflow, starting with C-based functional verification followed by Verilog RTL design for FPGA realization

8-bit model CPU - RTL design and Timing Analysis (Vivado).

Oct 2025 - Dec 2025

Guide : Prof. Ankesh Jain | Indian Institute of Technology Delhi.

- Designed an 8-bit model CPU in Verilog with instruction fetch from 256-byte memory, arithmetic (add/sub), logical shifts, and unsigned 8-bit multiplication.
- Performed static timing analysis to determine maximum operating frequency.
- Simulated functional correctness with a comprehensive testbench and synthesized the design using Xilinx Vivado.
- Reduced critical-path latency by modular RTL design.

Wi-Fi-Based Home Automation (Self Project)

Jul 2023 - Aug 2023

- Used NodeMCU to remotely control electrical appliances.
- Used BLYNK app for controlling relay operation via a smartphone.

Demonstration of Vertical Axis Wind Turbine for Highways

Jan 2022 - Jun 2022

Guide : Asst. Prof. Abhijeet Lal | Bhilai Institute of Technology Durg.

- Developed a vertical axis wind turbine to harness wind turbulence from moving vehicles on highways.
- Connected the turbine to a DC motor with a 1:10 gear ratio and a battery charging circuit.
- Successfully demonstrated a renewable energy solution for power generation.

Skills

Languages: Verilog | C++ | Python (Beginner) | MATLAB | HTML.

Software: Vivado | LT Spice | Microsoft Excel | Microsoft Word | SCCM | Service Now.

Courses Done: CMOS Digital IC Design | Digital Communication | Signal Processing Lab | Digital Electronics.

Ongoing Courses: Semiconductor Memory Design | MIMO Wireless Communication.

Achievements

- 1 Secured **97.06 percentile** in **GATE EE 2025** among candidates who appeared for the examination.
- 2 Received appreciation for **outstanding contribution** during **JFM-23 Quarter** at **HCL Tech**.
- 3 **Second position** in Academics in **Class XII**.